

Short Analysis Pipeline

BIOLM0051

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Scenario

A suitcase containing deez several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

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Introduction

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Methods

Pipeline Overview

The FASTQ files in this report are small and the use of HPC resources in this case is not necessary. However, this workflow is designed to scale easily to similar analyses with much larger datasets. All steps here are able to run on an HPC cluster without modification, demonstrating reproducibility, automation, and capacity for high-throughput analyses.

Note: In addition to the part-wise analysis (e.g., A1, A2, A3), an exploratory alignment and tree inference were also performed using the merged FASTA sequences for each sample (A, B, C, D combined). This reruns scripts 9–11 but is not part of the core pipeline.

Some python code:

Some Bash script:

Table 1: Top 15 BLASTP hits for sample OMRGCv2f.11922010.1 against SwissProt.

Query	Subject	% Identity	Alignment Length	E-value	Bitscore
OMRGCv2f.11922010.1	A5UYW4.1	55.56	63	1.3e-17	85.1
OMRGCv2f.11922010.1	A7NS65.1	55.56	63	1.8e-17	83.5

Results

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Alignment results using Clustal Omega on some test data:

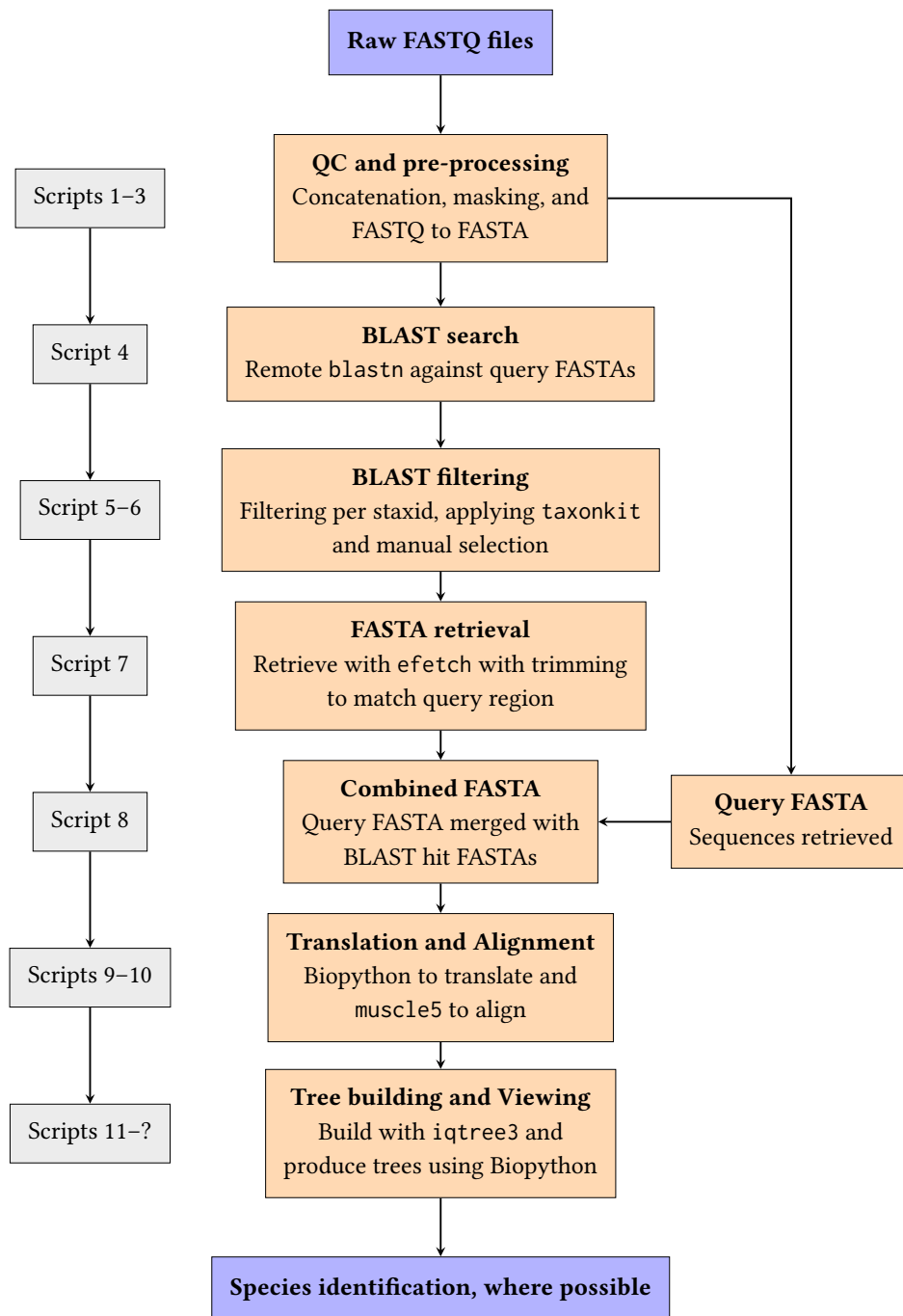


Figure 1: Workflow of analysis pipeline.

Discussion

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References

- [1] WJ Kent. Blat—the blast-like alignment tool. *Genome research*, 12(4):656–664, 2002.
- [2] MJ Wolin, TL Miller, and CS Stewart. Microbe-microbe interactions. In *The rumen microbial ecosystem*, pages 467–491. Springer, 1997.
- [3] B Baker, P Zambryski, B Staskawicz, and SP Dinesh-Kumar. Signaling in plant-microbe interactions. *Science*, 276(5313):726–733, 1997.

Acknowledgments

This work was carried out using the computational facilities of the Advanced Computing Research Centre, University of Bristol - <http://www.bristol.ac.uk/acrc/>

Supplementary Data

Supplementary data and scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git